University of Puerto Rico, Mayaguez University Campus
Department of Electrical and Computer Engineering

CSP Simplified Assignment

by
Diego H. Merchan Rueda
Marco A. Monserrat Montoto
Edgard A. Díaz Bobé
Malik J. Paramo Rios

For: Professor Jose F. Vega Riveros
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Statement of the Problem

The Einstein Riddle is a logic based reasoning problem that illustrates how structured constraints can lead to a single solution. This describes five houses, in which each has different attributes: colors, drinks, candies, pets and nationality[1]. The challenge of this problem is to determine who owns the zebra and who drinks water, based only on logical deductions.

This puzzle is modeled as a Constraint Satisfaction Problem(CSP), where each clue acts as a constraint that restricts the possible combinations of attributes. The solution of the problem emerges when all of the constraints are satisfied at the same time, to demonstrate the consistency of the solution. Aside from its logical side, the puzzle reflects how intelligent systems reason through structured data, balancing multiple variables and relationships in a coherent manner[1].

Review of Representations

Representation 1: Position based model

This representation treats each attribute as a variable whose domain is the set of house positions(from 1 to 5). Each clue becomes a constraint linking the variables through equality, adjacency or arithmetic. In the sentence "The green house is immediately to the right of the ivory house" can translate to [Green = Ivory + 1], and the sentence "The man who eats Hershey bars lives next to the man with the fox" becomes [Hershey - Fox = 1].

The advantages of the Position based representation lies in its compactness and symmetry. Every variable shares the same domain, and global *ALLDiff* constraints ensure that no attributes of the same type occupy the same house. This structure allows for solvers to apply constraint propagation and backtracking in an efficient manner, pruning the inconsistencies. This representation is ideal for demonstrating how CSPs can solve through domain reduction and consistency enforcement rather than brute force search[1].

Another one of its strengths is how it integrates with algorithmic reasoning and solver design in a natural way. This is because each variable is numeric, adjacency and ordering constraint can be expressed as simple arithmetic operations, which makes it more compatible with linear and integer programming formulations. This structure also supports the heuristic search strategies, like minimum remaining values(MRV) and least constraining value(LCV), allowing solvers to prioritize variables that are more restricted[2]. In practice, the position based model not only captures the puzzles's logic, but it also demonstrates how abstract reasoning can be optimized in a computational way, a bridge between human deduction and machine deduction.

However, the approach can feel abstract from the perspective of a human. The houses themselves are implicit, represented only in position, so the reasoning between neighbors or the visual adjacency requires translating numeric relations into mental images. It's the same logic used in SAT solvers and spreadsheet optimization models, where each variable’s position encodes the structural relationship between them. Overall, the position based model is efficient and elegant, but less intuitive for manual reasoning.

Representation 2: House based model

The house based model representation gives a different perspective from the last representation. Instead of focusing on positions, it instead centers on each house as a unit of reasoning. Every house is modeled as a tuple of attributes(color, nationality, drink, candy, and pet), while constraints are expressed as logical implications. An example of this model would be “if the house is yellow, then the candy is Kit Kat”. This approach aligns with how humans solve problems and puzzles. It’s a grid based reasoning method which is described in logic puzzle guides, in which solvers draw a 5x5 table and mark possible or impossible combinations[3]. Each elimination represents progress, and adjacency clues are handled naturally by comparing neighboring houses. The house representation is visual and conceptually rich, mirroring how humans think about spatial relationships and exclusivity.

Another aspect of the house based model is its interpretability and pedagogical value. Because each house is explicit, the reasoning process can be visualized for each step, making it ideal for explaining CSPs in an educational setting[4]. This also supports the incremental constraint addition, where new clues can be introduced in a dynamic manner without restructuring the entire model. This flexibility mirrors how humans can refine new hypotheses as new information is made available. For another part, from a computational standpoint, it is heavier since its logical disjunctions and neighbor constraints introduce complexity. Yet this representation is ideal for conceptual clarity and teaching, since it can show how CSPs can model real world reasoning. Overall, the house model is perfect for explanation and manual deduction, a great fit for educational purposes.

Representation 3: Permutation(or boolean) based model

If the first two representations were about how we see the puzzle, then this one is about how machines think. The permutation based model transforms the Puzzle into a web of binary decisions, instead of assigning each attribute to a house directly, it encodes every possibility as a Boolean variable{True if a value belongs to a house, false if not). It’s like a huge matrix where each cell represents a Yes-No question(such as “is the

Englishman in House 3, or "Is the zebra is House 2") and the solver's job is to make those align with the truth.

This approach is systematic and exhaustive, using 125 binary variables{5 houses x 5 attributes x 5 values), and every clue becomes a constraint that ties these variables together. The solver doesn't reason with words or colors, it reasons in bits. What makes this representation interesting is its compatibility with modern solvers. It can map directly to SAT and ILP formulations, meaning it can be handled by high performance engines like GoogleOr tools or Z3[5]. These solvers exploit unit propagation and conflict driven clause learning, pruning impossible combinations. In essence, the Boolean model turns the puzzle into a miniature version of how optimization problems are solved, from the scheduling to the circuit design[6].

The downside is its verbosity, since it has so many variables, the model becomes harder to read and explain. It's basically pure logic stripped of narrative and appearance, a version that speaks the language of the machine. Where the house mode feels intuitive and the position model feels elegant, the boolean representation is purely technical.

Analysis

After exploring the different representations of the Puzzle, it becomes clear that each approach reveals a distinct way of reasoning within CSPs. Each representation captures a different way of thinking, demonstrating three distinct perspectives of the logic itself. The house based model has a more natural, human feel, mirroring how we reason through our problems visually and relationally. The puzzle unfolds like a grid, where each clue narrows the possibilities until the pattern fits. Though the solver must handle more complex logical structures, the model grows heavier when automated.

The position based model, on the other hand, strips the puzzle down to its mathematical essence. It is clean and efficient, with every attribute becoming a variable, while its clue becomes a numeric relation. It is elegant in the simplicity, making it the most balanced option between human logic and computational precision. This representation captures the logic in a way that is both academic and practical. Meanwhile, the permutation model goes further into the computational territory. It does not care or understand the human values, only about truth values and logical consistency. It is technical, dense and powerful, the kind of representation that scales beyond puzzles into real optimization problems. Overall, the permutation is less about understanding, and more about performance. Beyond the puzzle itself, these representations illustrate how CSPs scale to real world applications such as scheduling, resource allocation, and circuit optimization.

While comparing them, the house based model wins in clarity, the permutation model wins in its computational depth, and the position based model sits in between both of these, structured yet intuitive. Taking into account their properties, the position based representation emerges as the best representation for solving the puzzle, since it captures the puzzle's logic with precision, while keeping it readable and structured. Unlike the Boolean model, it does not drown in variables, and unlike the house model, it doesn't rely on verbose logical implications. It's just enough for solver implementation, and clear enough for human interpretation. Ultimately, the Puzzle serves as more than a logic riddle, it shows how structured constraints can transform ambiguity into certainty, embodying the essence of constraint problems.

Reflection

Working on the problem gave us an idea of what representation really is in AI. By having a clear description of the problem, it was easier to build each model. The puzzle feels completely different depending on which lens we put on it. In the position based model the clues almost solved themselves once we wrote them as simple arithmetic. The house based version felt like sitting down with a pen and a 5x5 table, kind of like solving a logic puzzle in a magazine. And then the boolean model hit us with 125 variables and suddenly the whole thing looks way more intimidating than it actually was.

Something we didn't expect was how the most powerful representation, the boolean one, was also the hardest to actually read, while the house based model we initially dismissed as too informal ended up being the one we kept going back to when explaining our reasoning to each other. That tension between efficiency and explainability is a real issue in AI, and we ran into a small version of it ourselves. Working as a team made it more obvious, since each of us leaned toward a different model at first, and having just to justify our choices instead of just trusting our intuition. In the end what stuck with us the most was realizing that something that looks like a kids riddle is built on the same ideas behind scheduling, resource allocation, and circuit design, which made the whole assignment feel less like homework and more like a preview of the kind of problems we actually deal with.

Conclusions

No single representation is universally superior, the best choice depends on the context and goals presented. The position based model was exemplary in manual reasoning and algorithmic implementation, due to how it integrates with heuristic strategies. Giving it an edge in human deduction and general puzzles. For the house based model while computationally heavy it offered solid conceptual clarity and interpretability, which is good for step by step solutions that require visualization. As for the permutation based model it is mathematically concise, leveraged global constraints and mapped directly to SAT and ILP formulations, which is well suited for dedicated CP solvers. In practice, the position based model is the most balanced, while the house based model showed more capability in human interpretability. The permutation model showed promise when computational performance and scalability are the goal. All of them are not directly competing, but instead follow different perspectives of the same problem, each bridging human reasoning with computer intelligence.

References

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